

Keep On Track: Element D

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From the jump, our team had been inspired to build a marble wall that would be installed in an open cubie in the LifeSkill's sensory space. Based on previous devices in the classroom's sensory space, and immediate positive feedback from stakeholders, we began using this as our blueprint from the start. Because of this early decision in the design making process, there is very little divergence from our choice in assistive technology.

Marble Track Wall – Design Requirements

1. Purpose and Goals

- The marble track wall should provide interactive entertainment that encourages sensory engagement, fine motor skills, and cause-and-effect learning.
- It should be safe, calming, and inclusive for students with a range of physical and cognitive abilities.
- Be durable enough so kids can really beat on it and it still functions.

2. Safety and Accessibility

- No small removable parts that could be a choking hazard
- Smooth, rounded edges on all materials to prevent injuries.
- Constructed from non-toxic, durable materials (e.g., finished wood, acrylic, or high-grade plastic).
- Wheelchair-accessible height (bottom section no higher than 24" from the ground).
- All components should be securely attached to prevent detachment or falling.
- Easy to wipe and disinfect, with minimal crevices for dirt buildup.

3. Dimensions and Structure

- Wall area: approximately 3 feet wide and 3 feet tall.
- Bolted straight to the wall or floor so kids will not be able to move it.

4. Functionality and Interaction

- Tracks should allow marbles to roll smoothly from top to bottom without obstruction.
- Include interchangeable track pieces (ramps, funnels, spinners, chutes, bridges).
- Incorporate sensory features:
 - Visual: bright colors, transparent tubes, and motion.
 - Auditory: gentle clicking, rolling, or chime sounds.
 - Tactile: different textures on surfaces and handles.
- Magnetic or peg-based attachment system for easy rearrangement.

5. Educational and Therapeutic Features

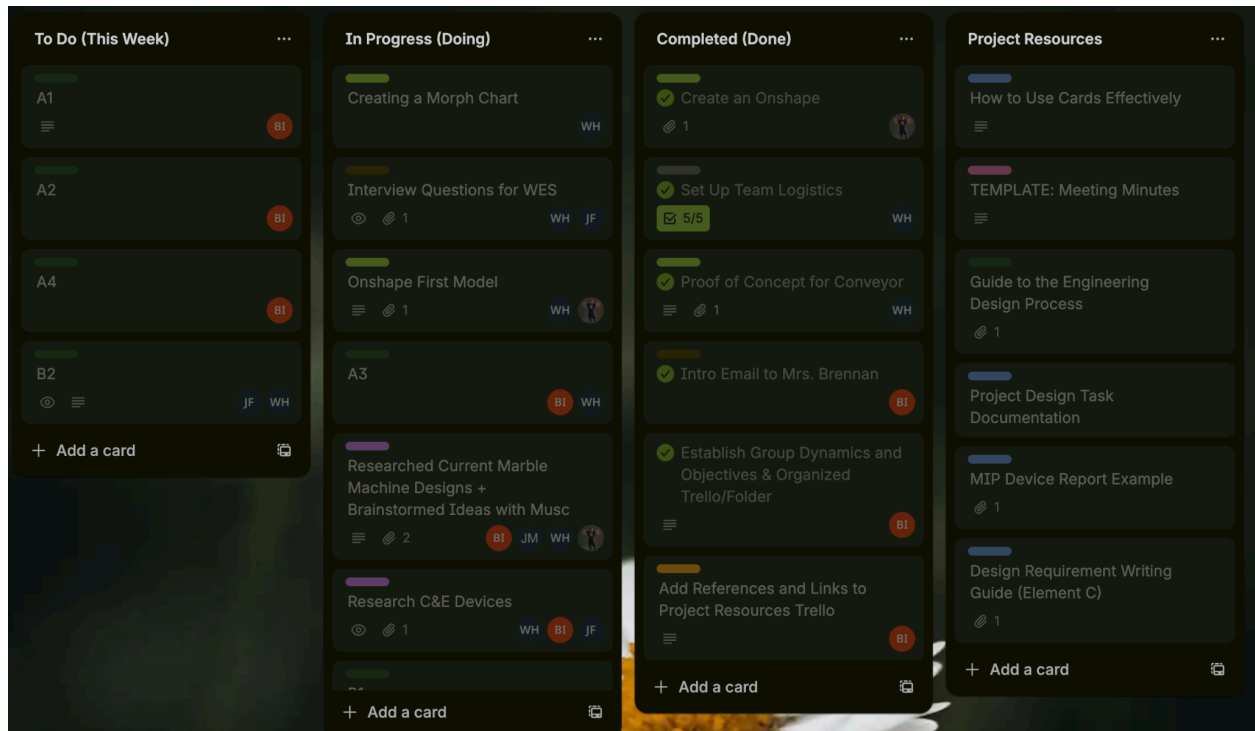
- Supports fine motor development (attaching tracks, releasing marbles, moving pieces).
- Optionally include light-up or sound-trigger features when marbles pass certain points.
- Allows for group play to encourage social interaction and teamwork.

6. Maintenance and Durability

- Must withstand daily classroom use and frequent touching.
- Components should be easy to replace if damaged.
- Use washable, water-resistant materials.
- A manual/guide for instructors if anything were to break

7. Aesthetics

- Bright but non-overstimulating colors (e.g., pastel tones or nature-inspired palette).
- Visually appealing design that complements classroom decor.



Our Trello has allowed the group to maintain an organized setting as we all manage our designated assignments. Since we have four members, two focused on the initial research part of developing assistive technology. The other two groupmates got handed the responsibility of curating inspiration for the marble wall.

Decision Matrix

Decision Matrix	
QODs	Weighted

Accessible	25%
Aesthetic	5%
Durable	10%
Engaging	15%
Inexpensive	10%
Safe	15%
Sensory	20%
Total Score	100%

After reaccessing the main priorities of the marble wall, our group narrowed down seven of the top functions we desire out of our prototype— all weighted on a scale out of 100%. First on the table is the most critical criteria: our group main goal is for the marble wall to be accessible. There is a wide range of unique needs and learning styles of the children in Whitemarsh’s Life Skills class. To weigh accessibility as 25% demonstrates our urgency to create a product that doesn’t exclude any child.

If you look around a playroom, the colors of the rainbow will be most prevalent. Toys should not be bland in color or bare in design. While aesthetics have a low weighting, our team still wants to hold it at a certain standard. There is no need to go

overboard with the colors as the marble wall serves as a therapeutic device but there is no harm in pops of color. Color is crucial in early childhood education and especially for children with color blindness. So yes 5% is minimal but again, much to be considered.

Durable is a category that we included that is geared towards the longevity of the device for the sake of the many instructors that help in the class. The last thing our group wants is a phone call in a year or two, with the teachers asking how to fix certain bits that they themselves don’t possess the skills to fix alone. The materials we use and the accuracy of our crafting is how we ensure durability. A weighting of 10% is a perfect fit and the group is keen to have the marble wall last for many years to follow.

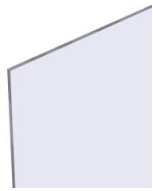
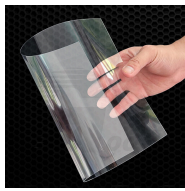
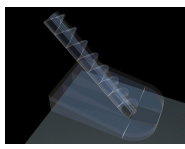
Engagement for the marble wall has two types. The first type of engagement is kick-back and relax, then the second type is more immersive: the face shoved against the TV sort of engagement. With this category weighted at 15%, the team deems it necessary to provide children with both types of engagement. Having a simple choice of the style of play a child wants teaches independence while teaching motor skills all at the same time. Engagement goes beyond the physical device and focuses more on the specific needs of an individual child.

There is no need to spend hundreds of dollars on a device that teaches basic developmental skills for children with disabilities. This expensive factor which is prevalent in many devices for Life Skills children creates much exclusion. Educational devices should be inexpensive so that all children no matter who, what, where, can learn important skills. With this reasoning, our group weighed inexpensive at 10%.

One of the first pieces of information Mrs. Brennan told us about her class was that some don't play gentle. The safety of the children, with no doubt, is one of the top requirements of the marble wall. There can not be choking hazards, broken glass/plexi, accessible wires and/or circuits, no wood splinters, etc. While this holds true to any toy dealt by children, it's even more of a safety issue when dealing with children with unique disabilities. Safety is rated 15% due to the environment being very abnormal and at times, very uncertain.

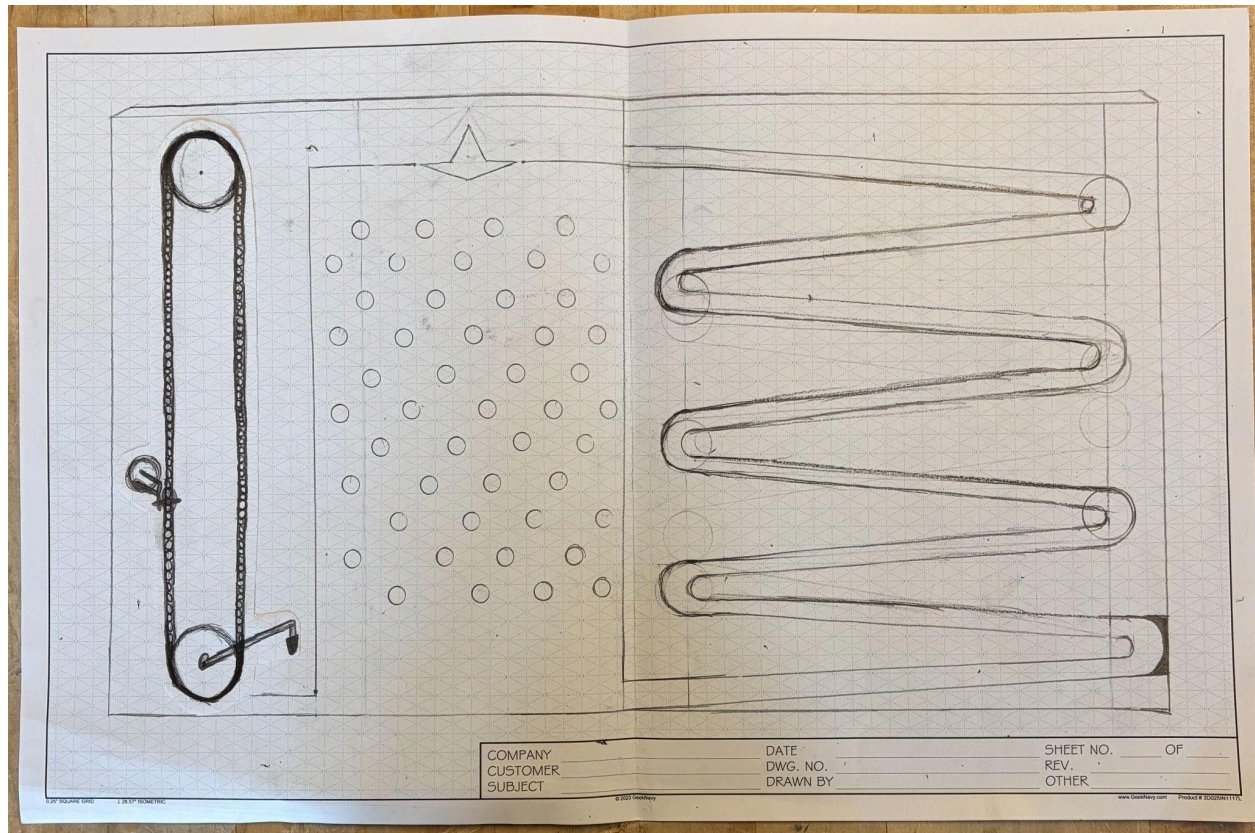
For our last category, sensory is weighted at 20%. The aspect of sensory is the marble wall itself— a device that includes color, sound, movement, and uncertainty. The perception of sensory mechanics is what we want the kids to identify when interacting with the marble wall. This is where the motor-skill development comes into play and this device will ensure that goal.

Marble Wall Morphological Chart:

Function <i>The Design Must...</i>	Concept 1	Concept 2	Concept 3	Concept 4	Concept 5
Withstand constant abuse	¼ inch plexi 	½ inch glass 	Clear plastic 	Bulletproof glass 	Tempered glass 
No clogging/back up	No intersections 	Wider tack 	Larger clearance 	Vibrating unit 	ball/track material 
Bring marbles to top	Motor 	Hand crank 	Spiral conveyor 	Lifting table 	Pinball method 

					
Motor Skills Development	Hand crank 	Hold back slide	Manual elevator	Pinball 	On /Off buttons 
Mounting strength	Molly bolt 	Cinder block screw 	Concrete screw 	Flex seal glue 	Toggle bolt 
Marbles	Steel 	Glass 	Plastic 	Rubber 	Ceramic

Concept Drawing



This drawing is a mock-up of the vision for the marble wall. To the left is the hand-crank that will bring the marbles up to the top to be sent down either the plinko or the track— these two design choices are depicted on the right. The idea behind the plinko and the zig-zag track is for the inclusion of collaborative play but also to drill in cause-and-effect. The notion of cause-and-effect is displayed in the turning mechanism that will tilt the marble to motion further down the track to the zig-zag, or it will drop the marble down the plinko. The turning mechanism will teach the child that there is a reaction for every action. This is a critical skill for the Lifeskills student to learn as they are not born with the same social cues as others.

Down below are documentations from Onshape. The first picture is the drawing file for the prototype. Then the second picture is an isometric view of our “Degree Testing” device. Before the initial prototype, it was important to test out the degree of the track—see which one is optimal for consistent speed. The mock-up tested out 15 degrees and 8 degrees, 8 being the best choice of the two.

Onshape CADD Drawings:

